

# PakkaHisaab (पक्का हिसाब)

पाँच रास्ते, एक सच — Multimodal AI Cashbook & GST Protection for Bharat Microbusinesses

[illegible]

Made with **GAMMA**

## THE PROBLEM

# 63 Million Indian Microbusinesses Trapped in Informal Record-Keeping

From the kirana store to the local hardware trader, Bharat's backbone runs on paper, memory, and trust. This informality creates invisible losses and legal exposure that no shopkeeper can afford.



### Scattered Financial Records

Handwritten notebooks, paper bills, and WhatsApp UPI screenshots lead to lost entries and uncollected debts. There is no single source of truth — just fragmented memory across multiple mediums.



### Manual Arithmetic Losses

Human math errors in daily ledgers cause unseen losses of **₹10,000+ per trader annually**. A misplaced decimal or missed entry compounds silently over months.



### GST Notice Vulnerability

Lack of digital audit trails leaves shopkeepers helpless against tax mismatch notices and heavy penalties. Without structured records, responding to a GST notice becomes nearly impossible.

THE SOLUTION

# "AI Reads & Reasons; Only Code Touches the Math."

PakkaHisaab digitizes multimodal inputs, reconciles entries deterministically, and protects the owner against GST notices — all without changing how a Kirana owner naturally works.



## 100% Integer Paise Math

Our Python accounting engine computes all totals in **integer paise** — zero floating-point errors, zero LLM calls touching math. Every rupee is accounted for with surgical precision.

**0 Floating Errors**

**0 LLM Math Calls**



## Evidence Passport

Every single ledger number links directly to its source file, bounding area, confidence score, and dual-language (Hindi/English) proof. Click any row — see the exact document behind it.

**Source File Linked**

**Bilingual Proof**

# 5 Inputs — Zero Change in Kirana Owner Behavior

Shopkeepers don't need to learn new workflows. PakkaHisaab meets them exactly where they are — with the tools and habits they already use every day.



## Khaata Photos

Handwritten mixed Hindi/English notebooks digitized via Vision AI (gpt-4o). No retyping required.



## Indic Voice Notes

Speak Hindi/Hinglish ("रमेश को ₹2500 दिए") via Sarvam AI sovereign model (saaras:v3). Native number parsing included.



## Supplier Invoices

Structured photo extraction of taxes & line items from any supplier bill — GSTIN, HSN, and totals parsed automatically.



## Natural Language Chat

Query cashflow and party balances in Hindi or English. Ask naturally, get precise answers instantly.



## UPI & Bank Statements

Instant CSV & screenshot parsing with 1.0 confidence matching. Every transaction reconciled in seconds.

# Deterministic Reconciliation with Click-to-Verify Proof

## 5-Priority Matching Engine

Transactions are matched through a deterministic priority chain — from exact references down to voice-confirmed fuzzy matches. No guesswork, only verifiable logic.

01

### **exact\_ref**

Direct invoice or reference match

02

### **exact\_amount\_date**

Amount + date alignment

03

### **window ( $\pm 3$ days)**

Date proximity matching

04

### **fuzzy\_party ( $\geq 0.85$ )**

Similar party name match

05

### **voice\_confirmed**

Manual voice verification

## 4 Automated Anomaly Detectors

The system continuously scans for four categories of financial anomalies, flagging each with a clear explanation and source link.

### **Unmatched Invoices**

e.g., photo INV-231 for ₹4,800 without corresponding UPI payment

### **Duplicate Entries**

Same transaction recorded twice across different input channels

### **Khaata Math Errors**

Handwritten totals that don't match line-item sums

### **Personal vs. Business**

Ambiguous expenses flagged for owner review before GST filing

-  Click any ledger row to reveal its exact source document proof — photo, voice clip, or UPI screenshot — in the interactive drawer.

# GST Kavach — Tax Protection Shield for MSMEs

GST Kavach transforms a terrified shopkeeper into a prepared one. From proactive risk scoring to bilingual notice responses, every tool is designed for the Bharat trader facing the tax system.

## **Deterministic Risk Radar (0–100)**

A computed score combining three weighted factors: **GST Gap (60%)** for input-output mismatches, **Open Exceptions (25%)** for unresolved anomalies, and **Personal/Business Ambiguity (15%)** for unclear expense categorization. Red flag before the tax officer does.

## **Bilingual Notice Response Generator**

Generates official reply drafts to GST notices in both **formal legal English** and **clear Hindi**. The shopkeeper understands the response, and the tax officer accepts it.

## **CA Audit Pack**

One-click ZIP download containing a **PDF Evidence Passport** and **structured CSVs** — everything a tax officer or Chartered Accountant needs for a clean audit trail.

# Indian Sovereign AI (Sarvam) + OpenAI Dual Router

PakkaHisaab is built on a dual-model architecture that prioritizes Indian sovereign AI while maintaining OpenAI as a reliable fallback — with full cost transparency and live telemetry.

## 🇮🇳 Sarvam AI — Primary Route

Indian sovereign models handle the most India-specific tasks natively, with zero dependency on foreign infrastructure for core Bharat use cases.

- **saaras:v3**  
Indic speech-to-text for Hindi/Hinglish voice notes
- **bulbul:v3**  
Hindi voice response generation
- **Native Number Parser**  
"पच्चास सौ" → ₹2,500 — understood natively

## 🔄 OpenAI Fallback + Telemetry

When Sarvam is unavailable or a task requires broader vision capabilities, the system routes to OpenAI with strict safeguards.

- **gpt-4o**  
Vision extraction for khaata photos and invoices
- **Whisper**  
Speech fallback when Sarvam times out
- **Live Telemetry**  
30s timeouts, retries, and explicit INR/USD cost tracking logged per call



The dual-router ensures zero downtime for Bharat traders while keeping costs predictable and every AI call auditable.

# Production-Ready, Zero-Login Demo for Hackathon Judges

PakkaHisaab is not a concept — it is a working prototype with real data, real edge cases, and a tech stack built for scale. Load it instantly, explore 71 pre-seeded entries, and reproduce every edge case in under 2 minutes.

## 71

### Pre-Seed Ledger Entries

Realistic mixed-language transactions ready for instant demo exploration

## ₹12,851

### Pre-Loaded Cashbook Value

Across multiple parties, invoices, and UPI transactions

## 4

### Reproducible Edge Cases

Duplicate entries, unmatched invoices, khaata math errors, and personal expense ambiguity

## 63M

### Target Microbusinesses

Bharat traders who need financial clarity and GST protection today

### Tech Stack

Next.js 14

TailwindCSS

Framer Motion

FastAPI

Supabase

Sarvam AI

OpenAI

### Bharat Impact Vision

Empowering 63 million microbusinesses with **financial clarity**, **zero math errors**, and **GST protection** — one kirana at a time. PakkaHisaab is not just a hackathon project; it is infrastructure for informal India's formal future.

✓ पाँच रस्ते, एक सच. Five ways in, one truth out.